SE LAB 6

NAME: SANCHITA

CLASS: 5H

Repo: https://github.com/sanchitavg/L6_L7_05_blood_donation_screening

Code Coverage

```
(venv) SANCHITA ~/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6 % pytest CodeCoverageTest_blood_
donation_screening.py -v
============================================= test session starts ==============================================
platform darwin -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0 -- /Users/sanchitaganesan/Desktop/SEM5/SE/LAB/selab6/L6_
L7_05_blood_donation_screening/venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/sanchitaganesan/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6
collected 13 items

CodeCoverageTest_blood_donation_screening.py::test_single_accepted_donor PASSED                          [  7%]
CodeCoverageTest_blood_donation_screening.py::test_invalid_input_raises PASSED                           [ 15%]
CodeCoverageTest_blood_donation_screening.py::test_invalid_volume_raises PASSED                          [ 23%]
CodeCoverageTest_blood_donation_screening.py::test_unregistered_cohort_skipped PASSED                    [ 30%]
CodeCoverageTest_blood_donation_screening.py::test_underage_donor_deferred PASSED                        [ 38%]
CodeCoverageTest_blood_donation_screening.py::test_low_hemoglobin_standard_cohort PASSED                 [ 46%]
CodeCoverageTest_blood_donation_screening.py::test_low_hemoglobin_high_altitude_resets_total PASSED      [ 53%]
CodeCoverageTest_blood_donation_screening.py::test_high_hemoglobin_deferred PASSED                       [ 61%]
CodeCoverageTest_blood_donation_screening.py::test_fast_track_bonus PASSED                               [ 69%]
CodeCoverageTest_blood_donation_screening.py::test_high_deferral_status PASSED                           [ 76%]
CodeCoverageTest_blood_donation_screening.py::test_centrifuge_no_data PASSED                             [ 84%]
CodeCoverageTest_blood_donation_screening.py::test_centrifuge_abort_spin PASSED                          [ 92%]
CodeCoverageTest_blood_donation_screening.py::test_centrifuge_stable PASSED                              [100%]
```

I ran pytest -v on CodeCoverageTest_blood_donation_screening.py.
All 13 tests were collected and every one passed.
These tests exercise each statement of the blood donation screening application.

```
(venv) SANCHITA ~/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6 % coverage report -m
Name                                          Stmts   Miss  Cover   Missing
---------------------------------------------------------------------------
CodeCoverageTest_blood_donation_screening.py      44      0   100%
blood_donation_screening.py                       45      0   100%
---------------------------------------------------------------------------
TOTAL                                             89      0   100%
```

I ran coverage report -m after running the tests with coverage.
blood_donation_screening.py shows 45 statements with 0 missed, so line coverage is 100%.
This is above the 90% target, and the Missing column is empty.

```
(venv) SANCHITA ~/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6 % coverage run -m pytest CodeCoverageTest_blood_donation_screening.py -v && coverage report -m
============================= test session starts ==============================
platform darwin -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0 -- /Users/sanchitaganesan/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/sanchitaganesan/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6
collected 13 items

CodeCoverageTest_blood_donation_screening.py::test_single_accepted_donor PASSED [  7%]
CodeCoverageTest_blood_donation_screening.py::test_invalid_input_raises PASSED [ 15%]
CodeCoverageTest_blood_donation_screening.py::test_invalid_volume_raises PASSED [ 23%]
CodeCoverageTest_blood_donation_screening.py::test_unregistered_cohort_skipped PASSED [ 30%]
CodeCoverageTest_blood_donation_screening.py::test_underage_donor_deferred PASSED [ 38%]
CodeCoverageTest_blood_donation_screening.py::test_low_hemoglobin_standard_cohort PASSED [ 46%]
CodeCoverageTest_blood_donation_screening.py::test_low_hemoglobin_high_altitude_resets_total PASSED [ 53%]
CodeCoverageTest_blood_donation_screening.py::test_high_hemoglobin_deferred PASSED [ 61%]
CodeCoverageTest_blood_donation_screening.py::test_fast_track_bonus PASSED [ 69%]
CodeCoverageTest_blood_donation_screening.py::test_high_deferral_status PASSED [ 76%]
CodeCoverageTest_blood_donation_screening.py::test_centrifuge_no_data PASSED [ 84%]
CodeCoverageTest_blood_donation_screening.py::test_centrifuge_abort_spin PASSED [ 92%]
CodeCoverageTest_blood_donation_screening.py::test_centrifuge_stable PASSED [100%]

============================== 13 passed in 0.01s ==============================
Name                                              Stmts   Miss  Cover   Missing
-------------------------------------------------------------------------------
CodeCoverageTest_blood_donation_screening.py         44      0   100%
blood_donation_screening.py                          45      0   100%
-------------------------------------------------------------------------------
TOTAL                                                89      0   100%
```

I ran coverage run -m pytest with -v and then coverage report -m in one command.
It shows 13 passed in 0.01s followed by the coverage table.
The total is 100% line coverage, which confirms the tests pass and cover the code.

Branch Coverage

```
(venv) SANCHITA ~/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6 % pytest BranchCoverageTest_blood_donation_screening.py -v
============================= test session starts ==============================
platform darwin -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0 -- /Users/sanchitaganesan/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/sanchitaganesan/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6
collected 30 items

BranchCoverageTest_blood_donation_screening.py::test_not_a_list_raises PASSED [  3%]
BranchCoverageTest_blood_donation_screening.py::test_negative_volume_raises PASSED [  6%]
BranchCoverageTest_blood_donation_screening.py::test_empty_list_valid_gives_zero_yield PASSED [ 10%]
BranchCoverageTest_blood_donation_screening.py::test_registered_false_and_missing_key_are_skipped PASSED [ 13%]
BranchCoverageTest_blood_donation_screening.py::test_registered_cohort_without_optional_keys_uses_defaults PASSED [ 16%]
BranchCoverageTest_blood_donation_screening.py::test_age_weight_boundaries[16-65-0] PASSED [ 20%]
BranchCoverageTest_blood_donation_screening.py::test_age_weight_boundaries[17-65-450] PASSED [ 23%]
BranchCoverageTest_blood_donation_screening.py::test_age_weight_boundaries[25-49-0] PASSED [ 26%]
BranchCoverageTest_blood_donation_screening.py::test_age_weight_boundaries[25-50-450] PASSED [ 30%]
BranchCoverageTest_blood_donation_screening.py::test_age_weight_boundaries[16-49-0] PASSED [ 33%]
BranchCoverageTest_blood_donation_screening.py::test_missing_donor_fields_default_to_deferred PASSED [ 36%]
BranchCoverageTest_blood_donation_screening.py::test_hemoglobin_boundaries[12.4-0] PASSED [ 40%]
BranchCoverageTest_blood_donation_screening.py::test_hemoglobin_boundaries[12.5-450] PASSED [ 43%]
BranchCoverageTest_blood_donation_screening.py::test_hemoglobin_boundaries[18.0-450] PASSED [ 46%]
BranchCoverageTest_blood_donation_screening.py::test_hemoglobin_boundaries[18.1-0] PASSED [ 50%]
BranchCoverageTest_blood_donation_screening.py::test_low_hb_in_standard_cohort_does_not_break_loop PASSED [ 53%]
BranchCoverageTest_blood_donation_screening.py::test_low_hb_in_high_altitude_cohort_breaks_and_resets PASSED [ 56%]
BranchCoverageTest_blood_donation_screening.py::test_high_altitude_cohort_with_normal_hb_unaffected PASSED [ 60%]
BranchCoverageTest_blood_donation_screening.py::test_high_altitude_break_only_stops_that_cohort PASSED [ 63%]
BranchCoverageTest_blood_donation_screening.py::test_fast_track_true_and_false PASSED [ 66%]
BranchCoverageTest_blood_donation_screening.py::test_fast_track_on_empty_cohort_gives_accepted_bonus PASSED [ 70%]
BranchCoverageTest_blood_donation_screening.py::test_four_deferrals_not_high_deferral PASSED [ 73%]
BranchCoverageTest_blood_donation_screening.py::test_five_deferrals_high_deferral PASSED [ 76%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_none_and_empty PASSED [ 80%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_two_consecutive_drops_abort PASSED [ 83%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_single_drop_then_recovery_is_stable PASSED [ 86%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_boundary_3000_is_not_a_drop PASSED [ 90%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_empty_run_is_stable PASSED [ 93%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_streak_resets_between_runs PASSED [ 96%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_abort_in_later_run PASSED [100%]
```

I ran pytest -v on BranchCoverageTest_blood_donation_screening.py.
All 30 tests were collected and passed, including the parametrized boundary cases.
They cover age, weight, hemoglobin, deferral count and centrifuge RPM boundaries.

```
(venv) SANCHITA ~/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6 % coverage report -m
Name                                                Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------------------
BranchCoverageTest_blood_donation_screening.py         72      0      0      0   100%
blood_donation_screening.py                            45      0     32      0   100%
-----------------------------------------------------------------------------------------------
TOTAL                                                 117      0     32      0   100%
```

I ran coverage report -m after running the tests with the --branch flag.
blood_donation_screening.py shows 32 branches with 0 partial and 0 missed.
Branch coverage is 100%, above the 90% target.

```
(venv) SANCHITA ~/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6 % coverage run --branch -m pyte
t BranchCoverageTest_blood_donation_screening.py -v && coverage report -m
============================================ test session starts ============================================
platform darwin -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0 -- /Users/sanchitaganesan/Desktop/SEM5/SE/LAB/selab6/L6
L7_05_blood_donation_screening/venv/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/sanchitaganesan/Desktop/SEM5/SE/LAB/selab6/L6_L7_05_blood_donation_screening/Lab 6
collected 30 items

BranchCoverageTest_blood_donation_screening.py::test_not_a_list_raises PASSED                                [  3%]
BranchCoverageTest_blood_donation_screening.py::test_negative_volume_raises PASSED                           [  6%]
BranchCoverageTest_blood_donation_screening.py::test_empty_list_valid_gives_zero_yield PASSED                [ 10%]
BranchCoverageTest_blood_donation_screening.py::test_registered_false_and_missing_key_are_skipped PASSED     [ 13%]
BranchCoverageTest_blood_donation_screening.py::test_registered_cohort_without_optional_keys_uses_defaults PASSED [
6%]
BranchCoverageTest_blood_donation_screening.py::test_age_weight_boundaries[16-65-0] PASSED                   [ 20%]
BranchCoverageTest_blood_donation_screening.py::test_age_weight_boundaries[17-65-450] PASSED                 [ 23%]
BranchCoverageTest_blood_donation_screening.py::test_age_weight_boundaries[25-49-0] PASSED                   [ 26%]
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BranchCoverageTest_blood_donation_screening.py::test_hemoglobin_boundaries[12.5-450] PASSED                  [ 43%]
BranchCoverageTest_blood_donation_screening.py::test_hemoglobin_boundaries[18.0-450] PASSED                  [ 46%]
BranchCoverageTest_blood_donation_screening.py::test_hemoglobin_boundaries[18.1-0] PASSED                    [ 50%]
BranchCoverageTest_blood_donation_screening.py::test_low_hb_in_standard_cohort_does_not_break_loop PASSED    [ 53%]
BranchCoverageTest_blood_donation_screening.py::test_low_hb_in_high_altitude_cohort_breaks_and_resets PASSED [ 56%]
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BranchCoverageTest_blood_donation_screening.py::test_fast_track_on_empty_cohort_gives_accepted_bonus PASSED  [ 70%]
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BranchCoverageTest_blood_donation_screening.py::test_five_deferrals_high_deferral PASSED                     [ 76%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_none_and_empty PASSED                        [ 80%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_two_consecutive_drops_abort PASSED           [ 83%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_single_drop_then_recovery_is_stable PASSED   [ 86%]
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BranchCoverageTest_blood_donation_screening.py::test_centrifuge_empty_run_is_stable PASSED                   [ 93%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_streak_resets_between_runs PASSED            [ 96%]
BranchCoverageTest_blood_donation_screening.py::test_centrifuge_abort_in_later_run PASSED                    [100%]

============================================ 30 passed in 0.02s =============================================
Name                                                Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------------------
BranchCoverageTest_blood_donation_screening.py         72      0      0      0   100%
blood_donation_screening.py                            45      0     32      0   100%
-----------------------------------------------------------------------------------------------
TOTAL                                                 117      0     32      0   100%
```

I ran coverage run --branch -m pytest with -v and then coverage report -m together.
It shows 30 passed in 0.02s followed by the branch coverage table.
Every True and False outcome of every decision was tested, giving 100% branch coverage.